

Module 3 Assignment

AI Workflow Analysis

Part 1: Workflow Analysis

Description: I maintain a Lambda function in AWS whose operation is to perform multiple API calls to gather information about vulnerabilities discovered in ECR images across multiple AWS accounts. The function produces an Excel report that is written to an S3 bucket (AWS object store). Periodically, a request to add an additional account is submitted through a Jira ticket.

Workflow:

1. Create a Jira ticket requesting a new AWS account be added.
2. Monitor email for notification of the Jira ticket.
3. Perform git pull to get the latest code update.
4. Open code in VS Code and update the appropriate lines.
5. Compile and deploy the package.
6. Trigger the new Lambda function.
7. Verify the report contains the new account.

Part 2: The Assessment

Step 1 – Create a Jira ticket requesting a new AWS account be added.

Jira AI Agent

- Has built-in natural language capabilities for creating Jira tickets.
- The Jira AI Agent license is included with the Jira licenses, some of which are free for small implementations.
- Data may be sent to Atlassian/Jira services depending on the license and configuration. Enterprise licenses with strict data protections exist, but they are expensive. Free licenses may not prevent data from being shared.

Microsoft 365 Copilot & Teams

- Has plugins that can add natural language capabilities for creating Jira tickets.

- Microsoft 365 Copilot is a paid add-on for eligible Microsoft 365 plans, with enterprise pricing commonly based on a per-user subscription.
- Data may be processed within Microsoft 365 services, and Microsoft 365 licenses have strict data privacy restrictions that protect a company's data from being shared without authorization.

Step 2 - Monitor email for notification of the Jira ticket.

Microsoft 365 Copilot & Teams

- Has built-in abilities to process Outlook emails and trigger agentic actions.
- Microsoft 365 Copilot is a paid add-on for eligible Microsoft 365 plans, with enterprise pricing commonly based on a per-user subscription.
- Data may be processed within Microsoft 365 services, and Microsoft 365 licenses have strict data privacy restrictions that protect a company's data from being shared without authorization.

Make AI Agents

- Can monitor email, parse Jira notification content, and trigger workflow actions across connected applications.
- Make has a free plan with limited monthly credits and paid plans that increase available automation credits and execution frequency.
- Data may be sent to a third-party automation platform because email and ticket data must pass through Make unless enterprise controls and connector permissions are carefully configured.

Step 3 - Perform git pull to get the latest code update.

Microsoft 365 Copilot

- Can assist with summarizing the Jira request and generating instructions, but it cannot directly perform a git pull without additional integrations to a development environment or automation tool.
- Microsoft 365 Copilot is a paid add-on for eligible Microsoft 365 plans, with enterprise pricing commonly based on a per-user subscription.
- Data may be processed within Microsoft 365 services, and care must be taken not to expose repository details or credentials through prompts or connected plugins.

GitHub Copilot

- Can assist in the development workflow by explaining git commands, helping troubleshoot pull errors, and working inside tools such as VS Code or GitHub environments.
- GitHub Copilot has free individual access with limits and paid business or enterprise licenses charged per user.
- Data may be sent to GitHub services because code context may be processed by GitHub, so repository access policies and data-sharing settings should be reviewed.

Step 4 - Open code in VS Code and update the appropriate lines.

VS Code Agent Harnesses

- Can use agent-based coding workflows in VS Code to inspect files, suggest edits, update configuration, and help modify the correct source code section.
- Cost depends on the specific agent extension or service used. Some extensions are free, while others require paid subscriptions or usage-based billing.
- Data may be sent to external AI services depending on the extension provider and model configuration because source code may be shared for analysis.

GitHub Copilot

- Can recommend code changes, generate snippets, explain existing logic, and assist with unit tests or refactoring inside VS Code.
- GitHub Copilot offers individual and organizational plans, with business and enterprise licensing charged per user.
- Data may be sent to GitHub services because source code context may be processed by GitHub, so enterprise policy controls should be enabled.

Step 5 - Compile and deploy the package.

VS Code Agent Harnesses

- Can help run build commands, interpret compiler errors, update deployment scripts, and guide package deployment from the development environment.
- Costs vary by the selected VS Code agent or extension, ranging from free tools to paid subscriptions or usage-based AI services.
- Data may be sent to an external provider if code, logs, or deployment output are shared during troubleshooting.

GitHub Copilot

- Can assist with build failures, suggest deployment command corrections, and help generate or update deployment scripts.
- GitHub Copilot business and enterprise plans require a paid per-user subscription, while limited individual options may be available separately.
- Data may be sent to GitHub services because logs and deployment files may include sensitive account information, so they should not be shared unless appropriate protections are in place.

Step 6 - Trigger the new Lambda function.

Amazon Q Developer

- Can help generate AWS CLI commands, explain Lambda invocation options, diagnose AWS console errors, and assist with troubleshooting execution issues.
- Amazon Q Developer has a free tier with monthly limits and a Pro tier billed per user for higher limits and administrative controls.

- Data may be processed within AWS services, and privacy controls are stronger for AWS-hosted workflows, but sensitive account identifiers and credentials still need to be protected.

Make AI

- Can automate a Lambda trigger by connecting approved workflow steps to AWS services or API calls after a Jira or email event occurs.
- Make includes a free plan with limited credits and paid plans that increase automation capacity and execution frequency.
- Data may be sent to Make because AWS account information, API credentials, or execution results may be routed through the platform, so least-privilege connectors and secrets management are important.

Step 7 - Verify the report contains the new account.

Amazon Q Developer

- Can help query S3, review Lambda output, generate AWS CLI commands, and troubleshoot whether the report includes the newly added account.
- Amazon Q Developer has a free tier with limited usage and a paid Pro tier for expanded limits and enterprise administration.
- Data may be processed within AWS services, and report contents may contain account or vulnerability information that should be handled as sensitive data.

Make AI

- Can automate verification by retrieving the Excel report from S3, checking whether the new AWS account is present, and notifying the requester when validation is complete.
- Make offers a free plan for light automation and paid plans based on monthly credits and team or enterprise features.
- Data may be sent to Make because the Excel report may contain vulnerability and account data, so enterprise controls, encrypted connectors, and restricted access should be required.

Part 3 Reflection

This workflow is well suited for automation, and AI tools can perform each of the steps, but there is not a single AI tool that can perform all the required steps. The request to add an account to the code in question is well defined, and AI tools could be useful in several parts of the process. However, breaking the process into several different AI components may make a straightforward task more complicated than required. Instead of multiple AI tools integrated across several steps, a single AI agent based on a platform like Microsoft Copilot agent would be more suitable. A single AI agent using natural language capabilities and programmatic API calls could fully automate this task. The question remains whether the task can or should be expanded to include more complex work. There is a degree of technical judgment required to modify the Lambda function for other tasks. AI may be able to support the original task without a human in the loop, but it would not be recommended to use AI to complete more complex

tasks without human assistance. The final responsibility for more complex code changes should remain with a person until evidence of AI's capability has been thoroughly tested.

The skills that could be easily replaced by AI include creating a Jira ticket based on a natural language prompt, automatically processing an email notification, checking code into and out of a source code repository, making basic code changes, troubleshooting, code deployment, testing, and validation. The skills that would be difficult to replace involve judgments that AI may not have enough information to make. These include determining whether the person requesting the change is allowed to make the request and whether the requested account is eligible to be included in the report. If these skills are replaced, some hands-on learning could be lost. However, the loss may be outweighed by the performance gain whether AI is used as an assistant or as a full replacement. The benefit is strongest when AI handles low-risk repetitive work while the human still reviews, understands, and approves final actions.

Outsourcing part of this work to AI could affect my human mastery of the code, depending on how completely AI replaces my involvement in the process. If AI is allowed to make changes without my understanding them, I would be less capable of correcting a problem if AI makes a mistake. I would also lack an understanding of the code changes over time without being in the loop. AI could mitigate this loss with detailed documentation, but not completely. If I used AI to assist with explaining the options, identifying ways to optimize the code, checking work against the Jira ticket, and documenting the outcome, I would be better able to maintain the code long term. I would also be in a much better position to train another person to manage the code in my absence. If AI functions like a technical assistant, then my performance is more likely to improve, making it less likely that I would be replaced altogether. This would provide a greater sense of control over my vocation and over the aspects of my work that I want to completely hand over to AI.

The use of AI could serve my workplace by improving efficiency and helping ensure that adding new AWS accounts to the vulnerability report can be managed by people who have limited technical skills. This would also simplify the process of training new people to manage the process. A conflict could arise if the process were automated without a human in the loop. There are compliance and security considerations that require human review, and if the AI makes a mistake, the penalties can be severe. There is also the concern that the process may become so automated that, when an error is introduced into the process, the expertise needed to correct it could be lost.

There are areas where AI should not currently be used and may not be appropriate in the future. AI should not independently approve complex code changes or make unsupervised deployments. These activities involve accountability, system availability, system access, and compliance decisions that require professional judgment. In the future, AI may become more capable, but it should still operate within clearly defined boundaries where humans remain responsible for security, privacy, and final approval. Outside of this specific workflow, there are other areas of work where I do not believe it is appropriate for AI to be used now, and perhaps not ever. These include areas where moral judgment needs to be made, especially decisions involving a person's legal rights. For example, I would not want AI to replace lawmakers or judges. AI can grease the bureaucratic wheels of government, but I would not want AI to make the final decision. I do not think AI should be the deciding factor in college admissions, job hiring, firing, or promotions. These areas directly impact people's well-being and livelihood, and I am not convinced AI can provide the moral judgment these decisions require.

Word Counts:

Part 1: 122

Part 2: 1025

Part 3: 839